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Magnetization properties of intrusive/extrusive rocks from East Maio (Republic of Cape Verde) and their geological implications

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Summary. The remanent magnetization of intrusive/extrusive rocks of the 'basement' complex of East Maio constitutes four components that define two different axes of magnetization, at around dec. 328, inc. 12 and dec. 007, inc. 14 respectively. In general, two or more components co-exist in separate specimens or sites but both axes are present most frequently in the normal sense. The NNW-striking axis, the *B*-axis, fits very well with the Upper Cretaceous polar wander path for Africa. It is consequently inferred that the major phase of sheet intrusions in Maio dates from this time, probably from the interval 90–70 Myr BP. Comparisons of the directional dispersions in the folded and unfolded states suggest that this injection phase post-dates the uplift of the Central Igneous Complex of the island. The second axis of magnetization, the *A*-axis, agrees very well with late Teritary–Quaternary palaeomagnetic data for Africa and the Canary Islands. The *A*-axis is therefore regarded as of secondary origin, being the consequence of a thermal/chemical overprint during the Miocene–Pliocene volcanism on the island. The occurrence of a 50–70 Myr long period of volcanic quiescence and erosion, between the termination of the early igneous activity (Upper Cretaceous) and the rejuvenated magmatism in Miocene/Pliocene time, is compatible with similar observations in the Canary Islands. In contrast to the palaeomagnetic conclusions, the K/Ar data only give ages around 10 Myr. The unusually young isotope dates are regarded as being due to an almost complete age resetting and are seen in conjuction with the overprinted magnetization. This explanation is further supported by the fact that K/Ar results of pillow lavas underlying Upper Jurassic limestones only give Tertiary ages.

Introduction

The Cape Verde Islands are composed dominantly of basic volcanic material of assumed late Tertiary age but on the island of Maio a Mesozoic 'basement complex', consisting of intrusive/extrusive rocks and a 300–400 m thick limestone sequence, covers a major area in

the southern half of the island. It was the discovery of these fossiliferous late Jurassic–Cretaceous beds, apparently the oldest sediments exposed on Atlantic islands, that prompted the early geological interest in Maio. Pillow lavas and tuff layers occur in association with the Mesozoic sediments. At the bottom of the exposed stratigraphic section, and just below the limestones which are of deep sea origin, the volcanics are clearly submarine, but in the upper part of the sedimentary pile the volcanism appears to have originated in a subaerial/shallow water environment (Stillman *et al.* 1982). In addition to this important geological information the relatively remote distance of the Cape Verde Archipelago from the West African continental platform places the islands in a nodal position with respect to study initial stages of crustal evolution in the Atlantic. Thus, the Cape Verde Rise, on which the islands are situated, extends westward from the continental rise of Mauritania/Senegal, reaching nearly the eastern flank of the Mid-Atlantic Ridge. Therefore, a close consideration of the tectonomagmatic history of the Cape Verde Islands might be of crucial importance for evaluating the early palaeomagnetic setting and deep basin development of the Central Atlantic. Based on palaeomagnetic results this paper deals with aspects of the Upper Mesozoic tectonomagmatic history of the island of Maio. An earlier palaeomagnetic study of the Cape Verde Archipelago (Watkins, Richardson & Mason 1968) included most of the islands but from Maio only four sites of Neogene volcanics were studied.

Outline of the geological history; geophysical sampling

The major knowledge of the stratigraphy and petrology of the island is based on studies by Serralheiro (1970), Rigassi (1972), de Paepe *et al.* (1974), Klerkx & de Paepe (1976) and Stillman *et al.* (1982 and private communication). In the present context the stratigraphic scheme of Stillman *et al.* is adopted. A simplified geological map based on work by the latter authors is given by Fig. 1. The oldest rock unit is the Batalha Formation, the base of which is covered by the sea, consisting of pillow lavas and hyaloclastics. The overlying Morro limestone formation was deposited in a quiet depositional environment above the carbonate compensation depth, with no influx of terrigenous material, during the time span from late Jurassic to mid-Cretaceous (Rigassi 1972, 1975). The quiet pelagic sedimentation ended in the Albian. It was followed by more rapidly deposited and heterogeneous sediments (*ca.* 90 m in thickness) named the Carqueijo Formation. The calcareous tuff layers in this formation increase in abundance and thickness upwards and the complete rock assemblage is regarded as evidence for a rapid shallowing associated with initiation of a shallow water or subaerial volcanism. Further evidence for a shallow depositional environment comes from the presence of well rounded clasts in the tuff layers of the overlying Coruja Formation, suggesting the emergence of the island, and boulders of ankaramitic rocks indicate that erosion of previously covered plutonic rocks had taken place (Stillman *et al.* 1982). It therefore appears reasonable to conclude that by Albian/Senomanian time the crust in the Maio region had experienced an uplift of about 3–4 km but a significant tilting had not yet taken place as the Coruja Formation tends to be conformable with the preceding strata.

The island forms a dome-like structure; from the central unroofed igneous complex, consisting of eruptive breccias and lavas intruded by essexite and syenite dykes and plugs that in turn are cut by small carbonatite intrusions, the sediments and associated volcanics dip radially away at angles varying from nearly vertical to as little as 30°. Probably due to erosion post-Coruja Cretaceous sediments are unknown in Maio. The Central Igneous Complex shows various stages of magmatic activity. The term 'basement complex' as referred to here comprises the Batalha Formation, the overlying Mesozoic limestone succession with intercalated volcanics (the Morro, Carqueijo and Coruja Formations), and the Central Igneous Complex.

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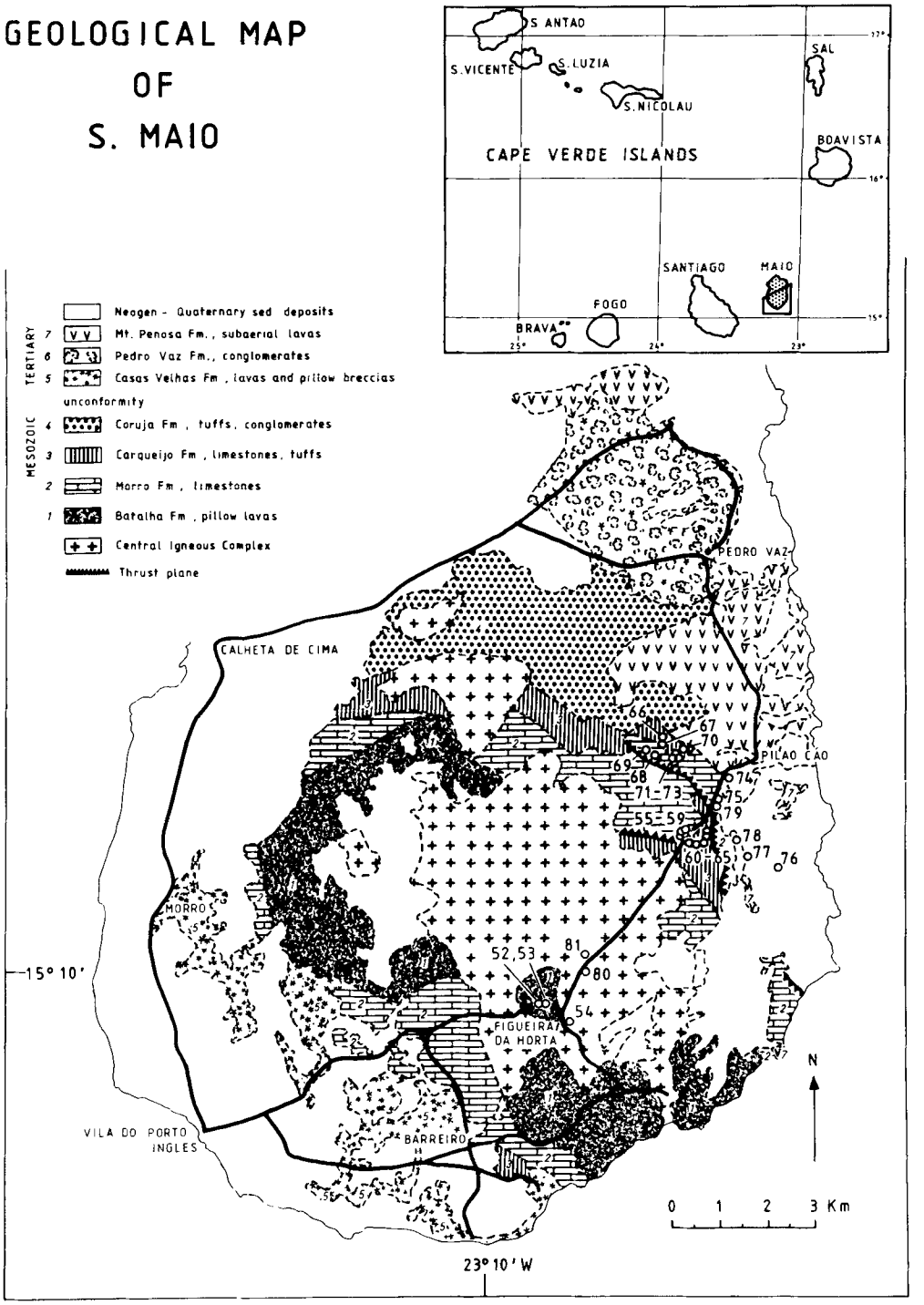


Figure 1. Geological map of South Maio, simplified after Stillman *et al.* (1982). Palaeomagnetic sampling sites are marked by open circles.

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The Mesozoic succession appears to have been planed off and overlain with marked unconformity by a Tertiary sequence of volcanics and sedimentary rocks. The lower part of this younger series, the Casas Velhas Formation, consists of lava deltas and subaerial ankaramitic flows which have been tentatively regarded as Palaeogene in age (Serralheiro 1970; Rigassi 1972; Stillman *et al.* 1982). Unconformably above these eruptives, and with material derived from them, is the diachronous Pedro Vaz Formation, consisting of fluvial conglomerates with tuff layers and some lava flows, from which Rigassi (1972) has reported mid-Miocene microfauna. The Pedro Vaz formation is overstepped by an extensive plateau of late Miocene–Pliocene lavas, particularly well exposed in northern and western areas, that rest on a gently dipping and locally lateritized surface. It is assumed that the original areal coverage of this basalt plateau has become much reduced by subsequent erosion.

Prior to the eruption of the Casas Velhas volcanics the Mesozoic (Upper Jurassic – *ca.* Cenomanian) rock formation has been extensively affected by sheet intrusions (Rigassi 1972; Stillman *et al.* 1982). It is possible that this sill and dyke complex, which may frequently have a density of between 50 and 80 per cent (i.e. 50–80 per cent of the host rock is obliterated) or more, represent different generations of magmatism; some may be connected with the volcanic activity within the Carqueijo and Coruja Formations, but the bulk of the intrusive activity is probably post-Coruja in age (see below). The major injection phase may have been associated with the doming process of the Central Igneous Complex, probably arising from high level plutonic emplacement. On the other hand, the recently discovered tectonic deformation on the eastern flank of the Central Igneous Complex, involving repetition of strata, is likely to pre-date the doming stage (Stillman *et al.* 1982).

The rock collection here concerned comprises mostly sills, intruding the Morro and Carqueijo Formations in East Maio. In some of these igneous strata Rigassi (1972) has noted pillow structures and Serralheiro (1970) appears to have been convinced about the existence of intercalated laval flows. However, there seems to be little doubt that the bulk of the sampled material is of intrusive origin; only in a few cases where the basaltic material is associated with thick calcareous tuff horizons do we suggest that the sampling represent lavas. Five sites were taken in vertical dykes, two of the sampled rock bodies have a greenish colour and others show macroscopically a more varying degree of alteration, but on the whole the field evidence suggest that the collected rocks are relatively fresh. In our palaeomagnetic collection from the Cape Verde Islands the Maio sampling locations are numbered 52–81. Of this material 29 sites, with a total of 110 oriented cores, represent the above-mentioned intrusive/extrusive rocks while one site (no. 74) was taken in a Miocene/Pliocene lava. For reference purposes an additional four flows of this late Tertiary volcanism, collected in the Island of Santiago, have been included in this study. All palaeomagnetic locations have been sampled by means of a portable drill, and sun compass orientations are available for *ca.* 35 per cent of the collected material. Total variation in declination estimates, based on the difference between magnetic and sun compass bearings, range between 11°W and 21°W, but the bulk of the data are close to the average value of 16°W. This is in very good agreement with the present regional declination for Maio (~ 15°W). Fig. 1 and Table 1 give further sampling details.

Analysis of the natural remanent magnetization (NRM)

PROCEDURES

A total of 118 specimens have been measured on a Digico Spinner Magnetometer (mostly a shielded version) and subjected to progressive demagnetization by means of alternating field and/or temperature. There is a relatively large proportion of low stability remanence in this

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Table 1. Details of palaeomagnetic sampling sites. See Fig. 1 for locations.

Site no.	Sample nos	Rock type and formation	Attitude of strata (strike/dip)
52	CV182–184	Vertical dyke, Batalha Fm	
53	185–187	Vertical dyke, Batalha Fm	
54	188–190	Vertical dyke, Central Ign. Complex	
55	191–196	Sill, Morro Fm	343/50 E
56	197–201	Sill, Morro Fm	350/40 E
57	202–205	Sill, Morro Fm	314/40 NE
58	206–210	Sill, Morro Fm	342/40 E
59	211–215	Sill, Morro Fm	343/50 E
60	216–218	Sill, Carqueijo Fm	322/52 E
61	219–220	Sill, Carqueijo Fm	309/52 NE
62	221–224	Sill, Carqueijo Fm	335/45 E
63	225–229	Sill, Carqueijo Fm	354/50 E
64	230–233	Sill, Carqueijo Fm	354/50 E
65	234–237	Sill, Carqueijo Fm	347/45 E
66	238–241	Lava?, Carqueijo Fm	290/40 N
67	242–244	Sill, Carqueijo Fm	284/45 N
68	245–248	Sill, Morro Fm	284/45 N
69	249–251	Sill, Morro Fm	325/54 NE
70	252–255	Lava w/tuff, Carqueijo Fm	324/55 NE
71	256–259	Lava?, Carqueijo Fm	329/60 NE
72	260–263	Lava?, Carqueijo Fm	Varying attitude: 260–262, 325/40 NE 263, 330/75 NE
73	264–267	Lava w/tuff, Carqueijo Fm	324/~ 60 NE
74	268–270	Noegen lava	~ horizontal
75	271–273	Sill, Carqueijo Fm	335–35 E
76	274–276	Sill, Carqueijo Fm	225/~ 60 SE
77	277–279	Sill, Carqueijo Fm	000/50 E
78	280–282	Lava w/tuff, Carqueijo Fm	324/35 NE
79	283–286	Sill, Carqueijo Fm	350/40 E
80	287–290	Vertical dyke, Central Ign. Complex	
81	291–294	Vertical dyke, Central Ign. Complex	

collection. For 28 of the tested specimens the signal/noise ratio decreased rapidly; erratic and non-reproducible results are encountered at a very early stage of demagnetization. These specimens are disregarded in the further discussion. Of the remaining 90 samples 60 define coherent directional groups by employing the following acceptance criteria: (1) Stable end point magnetization is accepted provided it can be experimentally confirmed by a minimum of three successive demagnetization steps (per specimen) associated with a steady intensity reduction. (2) Several samples show high NRM stability over the major range of intensity but terminating either in an erratic stage of behaviour or by initiating systematic directional trends at the last few steps of demagnetization. The stable range of this group of samples has been accepted as 'true' palaeomagnetic directions provided they represent at least 90 per cent of the NRM intensity. This procedure appears to be sound in that the latter category of results form coherent palaeomagnetic groupings, agreeing well with stable end point data from other specimens.

A greater number of samples show more or less consistent vectorial movement but with the intensity of magnetization decaying into the noise level before a terminal direction is reached. Some of the 'swinging' specimens (most of them shown in Figs 2–4) move along

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Table 2. Palaeomagnetic results giving characteristic directions of individual specimens (*cf.* text), the corresponding ranges of temperature or alternating field, and the palaeomagnetic group (*A* or *B*). All directions are without structural correction. vs denotes vector subtraction within specified temperature range.

Site	Specimen	*D*	*I*	Pal. group	Range
53	CV185-A1	357	22	*A*	NRM–550°
	186-A1	358	21	*A*	NRM–480°
	187-A1	354	24	*A*	NRM–21 mT + 50°–520°
54	188-A1	011	3	*A*	NRM–520°
	189-A1	002	–2	*A*	320–560°
	190-A1	006	5	*A*	100–540°
55	193-A2	015	34	*A*	120–21 mT + 50° and 100°
	194-A1	349	38	*A*	NRM–21 mT
	195-A1	011	3	*A*	340–420°
56	198-A1a	319	35	*B*	vs 100–250°
	198-A1b	199	4	*A*	250–480°
	199-A1	201	–13	*A*	200–420°
57	203-A1	001	29	*A*	13, 5–21 mT
	204-A1	357	48	*A*	NRM–15 mT + 100°
59	211-A1	355	20	*A*	NRM–27 mT + 100–350°
	213-A1	337	18	*B*	100–350°
	214-A1	000	14	*A*	9–27 mT + 140–350°
	215-A1	337	26	*B*	12–12 mT + 100–350°
62	223-A1	195	10	*A*	9–40 mT
	224-A1	199	2	*A*	NRM–21 mT
63	225-A1	326	14	*B*	NRM–12 mT
	228-A1	008	–7	*A*	NRM–450°
	228-A2a	005	–8	*A*	NRM–21 mT
	228-A2b	332	+3	*B*	vs (*cf.* Fig. 3)
	229-A1a	313	20	*B*	NRM–60 mT + vs (*cf.* Fig. 3)
	229-A1b	028	22	*A*	21–30 mT
64	230–A1	014	14	*A*	NRM–40 mT
	231-A1a	133	18	*B*	NRM–6 mT
	231-A1b	009	6	*A*	100–300°
	232-A1	003	24	*A*	NRM–22 mT
65	237-A1	176	–10	*A*	9–21 mT
66	239-A1	007	20	*A*	3–26 mT + 100° and 200°
	240-A1	355	24	*A*	3–14 mT
	241-A1	012	18	*A*	6–21 mT
68	245-A1	328	20	*B*	NRM–450°
	247-A1	010	–8	*A*	NRM–400°
	248-A1	359	2	*A*	490–525°
69	249-A1	321	22	*B*	520–580°
	249-A2	333	19	*B*	550–600°
	250-A1	335	21	*B*	350–500°
	250-A2	333	26	*B*	300–630°
	251-A1	326	18	*B*	350–600°
70	253-A2	313	16	*B*	100–500°
71	257-A2	007	10	*A*	100–425°
	258-A1	305	10	*B*	300–400°
	259-A1	328	23	*B*	150–525°
72	262-A1	010	30	*A*	NRM–500°
	263-A1	329	22	*B*	400–525°
	263-A2	328	2	*B*	460°, 480°, 500°
73	266-A1	026	3	*A*	100–300°

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Table 2 – continued

Site	Specimen	*D*	*I*	Pal. group	Range
75	271-A1	346	2	*B*	460–540°
	272-A2	335	5	*B*	300–500°
76	274-A1	327	12	*B*	200–525°
	275-A1	353	20	*A*	100–550°
78	281-A1	327	2	*B*	350°, 400°, 450°
	282-A1	342	–6	*B*	440°, 460°, 480°
79	284-A1	308	–10	*B*	NRM–250°
80	288-A1	004	2	*A*	NRM–525°
	289-A1	016	1	*A*	NRM–540°
	290-A1	019	44	*A*	NRM–560°
	290-A2	158	8	*B*	NRM–500°
81	293-A1	341	9	*B*	380–500°
	294-A1	001	32	*A*	100–525°
	294-A2	355	12	*A*	475°, 500°, 550°

well-defined great circle paths but these are numerically too few for an appropriate application of the remagnetization circle technique (Halls 1976). Vector subtraction analysis has been carried out on all specimens that define great circle trends on demagnetization. By this technique one can determine the direction of an erased vector provided there is a 'window' in the demagnetization spectrum for which only one component is being removed. When the various difference vectors cluster it is assumed that this basic requirement is fulfilled. A meaningful application of this technique would be to choose demagnetization steps that make the difference vector large enough to prevent undue influence by random measurement errors.

STRUCTURE OF THE PALAEOMAGNETIC RECORD

Figs 2–4 give a broad outline of demagnetization data, particularly for specimens that are thought to give true palaeomagnetic information (included in Table 2). The rocks are characterized by two shallow inclined axes of magnetization which are clearly discriminated by direction (see below). The most predominant one, called the *A*-axis, is striking NNE, while the second axis, the *B*-axis, has a NW–NNW direction. Both axes are represented by normal and reverse components. All four magnetization vectors may be present in a single site, and individual specimens may display various component combinations. It is frequently found, however, that the normal components predominate.

Fig. 2 illustrates cases of superimposed normal and reverse components. Specimen CV 231-A1 starts in the reverse *B*-axis position and remains there until the demagnetization field passes beyond 6 mT. At higher fields, and finally with three steps of thermal demagnetization (on top of the AF-treatment), the remanence vector performs a northerly movement terminating (at a low intensity level), in the normal *A*-axis direction. This end point magnetization agrees with the bulk remanence direction of two other specimens from the same site (CV 230-A1 and 232-A1). Note that *ca.* 92 per cent of the remanence intensity is linked with the initial NRM direction (reversed) so that the mean direction of the NRM–6 mT treatment steps has been accepted as a palaeomagnetic component (according to acceptance criteria). The normal magnetization probably represents less than 5 per cent of the total magnetization so that any directional adjustment of the NRM–6 mT direction is deemed unimportant.

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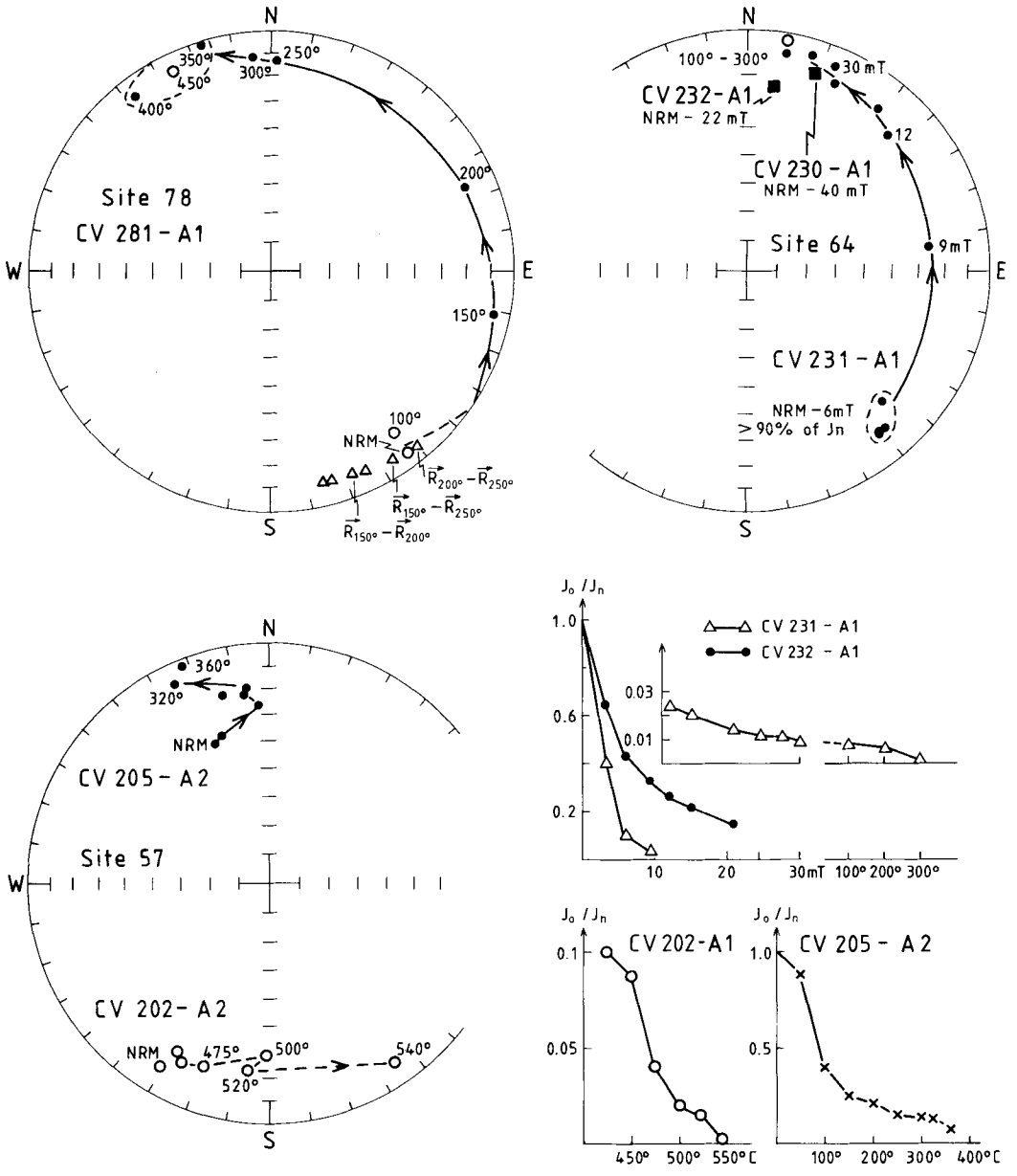


Figure 2. Demagnetization results (closed or open circles) of igneous rocks interstratified in the Cretaceous limestones of East Maio. Projection is equal area, and closed (open) symbols are downward (upward) pointing magnetization vectors. Thermal demagnetization steps are indicated by the temperature in °C while alternating field demagnetization steps are marked by the value of the peak field in mT. Square symbols are mean magnetization directions of closely spaced results obtained within demagnetization ranges as indicated. Vector subtracted directions are shown by triangles. Note the dual polarity and/or two axis build-up of the magnetizations in individual specimens or sites. The vector subtracted directions of specimen CV 281-A1 show a somewhat elongated distribution, suggesting a partial component overlap, and these results have therefore not been incorporated in the final palaeomagnetic data (Table 2). See text for further details.

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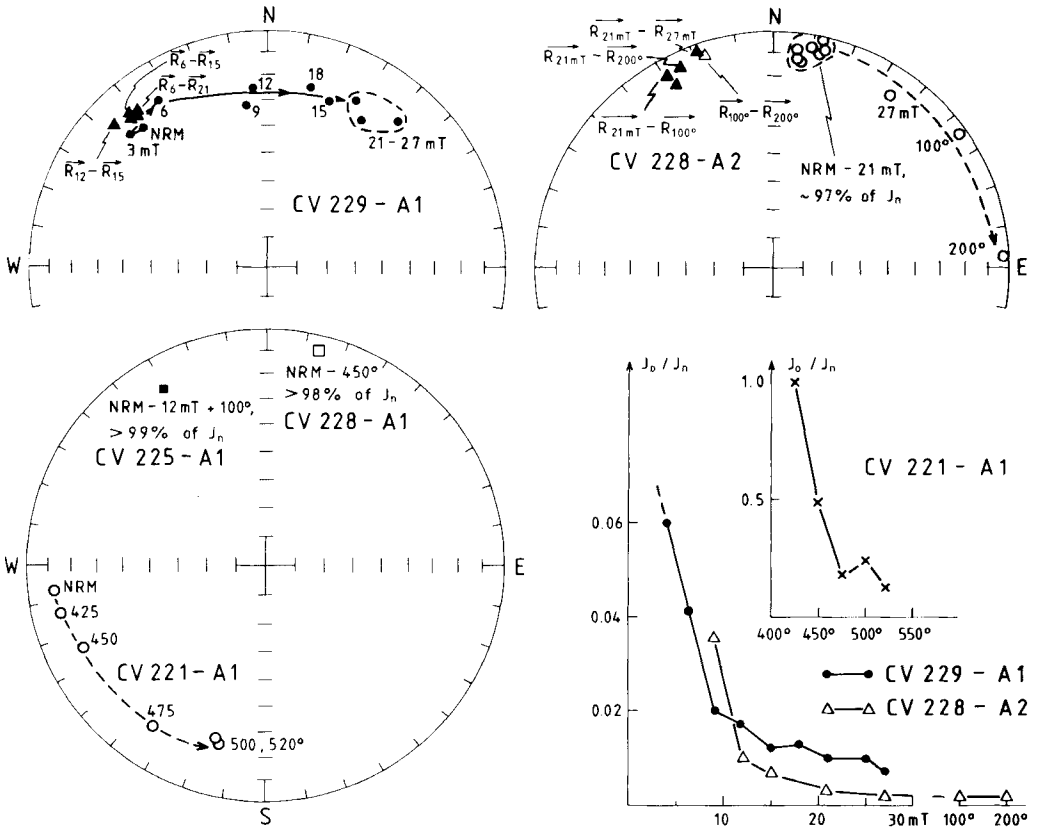


Figure 3. Further demagnetization data. Diagram conventions as for Fig. 2.

In this rock collection it is generally found that vector subtracted directions do not cluster but are spread out along the trace of the remagnetization circles. CV 229-A1 (Fig. 3) is one of the very few samples in this study for which the vector subtracted directions are well grouped, suggesting in this case that demagnetization in the 6–21 mT range only erases the *B*-axis component. It is not surprising that the vector subtracted magnetization corresponds so well to the remanence having coercivities $\leqslant 6$ mT since the latter magnetization tends to carry at least 95 per cent of the total remanence intensity. CV 228-A2 (Fig. 3) also responds well to vector subtraction and the normal components of both the *A* and *B* magnetization tend to be retrieved. It must be emphasized, however, that the *A*-direction of the latter specimen is unlikely to represent a completely clean magnetization. The overlap in stability ranges is a common problem in multicomponent palaeomagnetism and the apparent directional stability, covering some 97 per cent of the natural intensity, must not be taken as evidence for insignificant *B*-vector contribution. It is more realistic to assume that if the completely clean *A*- and *B*-vectors of specimen CV 228-A2 had been established the directional divergence between them would have been larger than presently demonstrated. Specimens like CV 202-A2 and 205-A2 (Fig. 2) suggest that the same two-axis magnetization is present but the results do not satisfy our acceptance criteria.

The data for the 60 samples that are considered to give relevant palaeomagnetic directions are listed in Table 2. For reasons given below all data are without dip correction. The same

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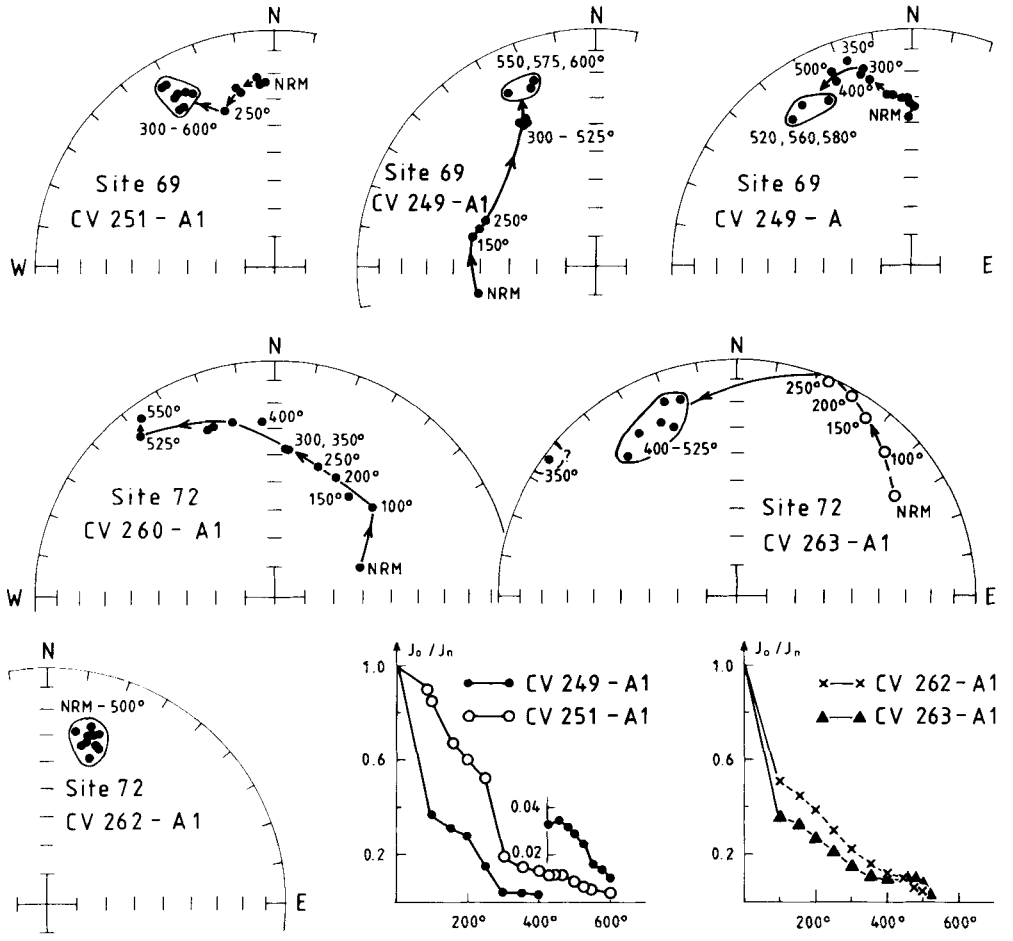


Figure 4. Further demagnetization data. Diagram conventions as for Fig. 2.

data are visualized in Fig. 5. The *A* and *B* groups do not overlap and the two mean directions are significantly different at the 95 per cent significance level (*cf.* Table 3). This discrimination between the mean directions is also very clear from Watson's Test (Watson 1956); the statistic value is 75.8 compared with a critical value of 19.5 from *f*-ratio tables ($p = 0.05$).

Of the 14 analysed specimens from the five investigated Miocene/Pliocene lavas nine specimens satisfy the reliability criteria employed in this study. These latter results, which are plotted as triangles in Fig. 5(c), agree extremely well with the *A*-axis magnetization obtained from the remaining rocks. Apart from varying low-stability components the late Tertiary lavas seem to have a single-component build-up of their remanences, while the older intrusive/extrusive rock sequence commonly possess multiphase magnetizations.

SOME MAGNETO-MINERALOGICAL RESULTS

The inferred multicomponent remanence, a magnetization structure that is very different from that expected if the rocks had dominantly retained their original cooling remanence, should somehow be reflected in texture and composition of the iron-titanium oxides.

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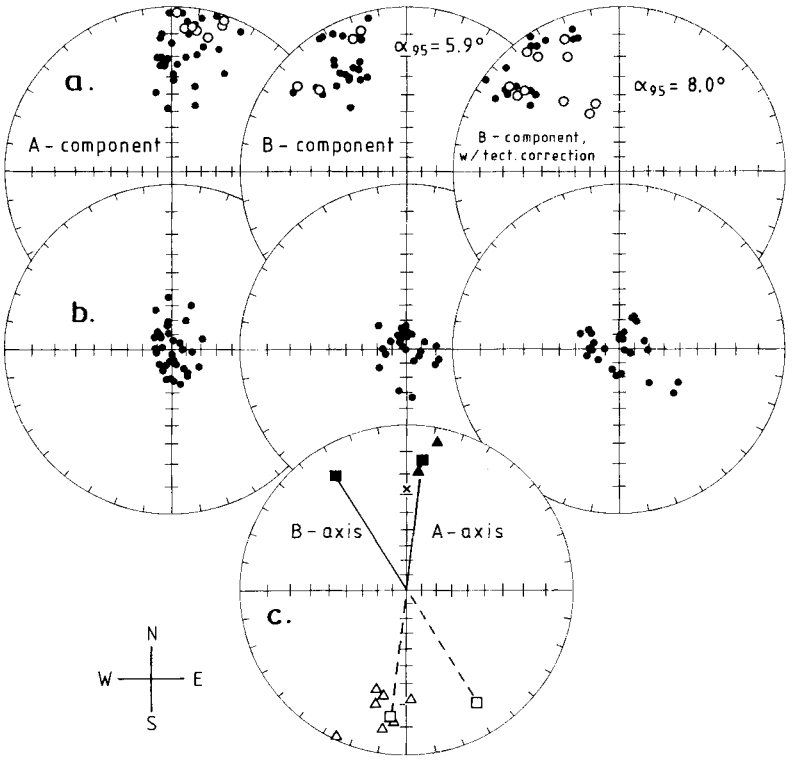


Figure 5. Palaeomagnetic groupings (a) based on individual specimen results. All reserved directions have been rotated into the normal sense so as to show the shape of the populations more clearly. (b) displays the various directional dispersions centred on each population (*A*- and *B*-axis data, the latter before and after structural correction). The 'N–S' axis of (b) lies in the plane through the origin and the mean directions of the relevant plots in (a). The *B*-axis of (c) is based on tectonically uncorrected data. The triangular symbols represent specimen results from a few Neogen basalts. See text for further details.

Reflection microscopy has so far only been carried out on a few samples, from site nos 54, 55, 71 and 78. A characteristic feature is the apparent absence of ilmenite lamellae in the titanomagnetite (TM) grains, suggesting that high temperature oxidation is at most only developed to a very modest extent. After cooling the homogeneous TM grains are assumed to have had low Curie points (T_c) related to a high titanium content. In all probability the

Table 3. Mean palaeomagnetic data from the old igneous complex of Maio.

Group	N	$\bar{D}$	$\bar{I}$	R	κ	α_{95}	Pole
A	38	006.7	+13.6	36.3	21.5	5.2	297.9 E, 80 S
B	26	327.8	+11.9	25	24	5.9	054.6 E, 57.1 S
$B^\star$	26	321.0	−2.0	24.1	13.5	8.0	047.9 E, 48.2 S

Data for the *A* and *B* groups are from Table 2, but with reversed directions inverted. The $B^\star$ group represents the *B*-data corrected for tectonic dip (*cf.* Table 1).

N: number of unit vectors (specimens).

R: length of resultant vector.

κ: precision parameter.

α_{95}: radius of circle of confidence at 95 per cent significance level.

$\bar{D}, \bar{I}$: declination and inclination of mean vector.

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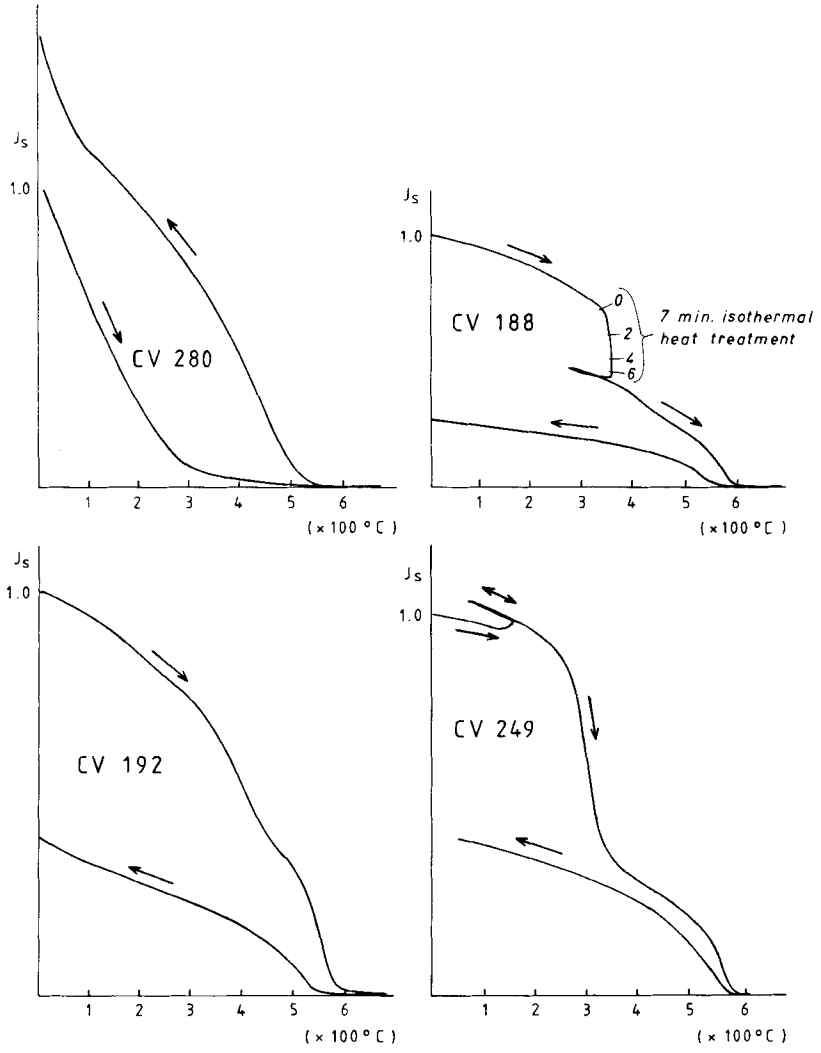


Figure 6. Examples of saturation magnetization versus temperature. See text for details.

original titanomagnetites were therefore fairly susceptible to low-temperature alteration, and microscopic analysis shows that such processes have indeed taken place. Thus, the colour of the euhedral TM particles show shades of brown, grey and white, along with displaying granulation textures. This suggests that maghemitization may be of great importance in these rocks in addition to probable conversion of titanomagnetite to magnetite through granulation processes. As demonstrated in Fig. 6 saturation magnetization versus temperature (J_s-T) measurements support the microscopic evidence. Sample CV 280 show a J_s-T behaviour that conforms to a relatively unaltered titanomagnetite but generally the 'Curie' points are much too high for unexolved basaltic TM grains. Another important factor is the rapid decay of saturation magnetization on isothermal heat treatment at moderate temperatures, *cf.* CV 188 Fig. 6, indicative of the break-down of maghaemite. To judge from the relatively low J_s after heat treatment one must assume that some haematite has formed through oxidation and/or mineral transformation. Sample CV 249 show a typical irreversible

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'kink' at around 150°C. This feature was originally described by Ade-Hall, Palmer & Hubbard (1971) and related to rocks that have undergone either high deuteric oxidation or high regional hydrothermal (low temperature) alteration. The thermal decay patterns of the natural remanent magnetization, *cf.* Fig. 4, have characteristic blocking temperature ranges that agree with the 'Curie' points as determined from J_s–T measurements (*ca.* 300°C and 550°C respectively). On the whole the mineralogical properties are fully compatible with the existence of a multiphase magnetization in these rocks.

Interpretation

The experimental data outlined above suggest that the basaltic rocks in the Mesozoic sediments of East Maio have experienced two magnetization 'pulses', each of them most likely involving lengthly acquisition processes. This probably implies that the palaeomagnetic record of geomagnetic secular variation is greatly suppressed so that the principal scatter of remanence directions may be caused by unresolved multicomponent magnetization, arising partly from the presence of a dual polarity structure in each of the magnetization groups (*A* or *B*), or the mutual interaction between the two axes of magnetization.

The overall palaeomagnetic results, including statistical parameters and pole locations, are listed in Table 3. As seen from Fig. 7 the *B*-magnetization, before structural unfolding, fits

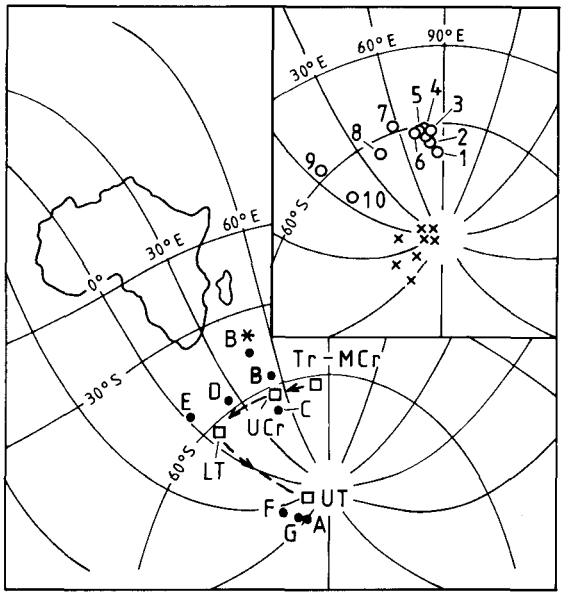


Figure 7. Pole position from the Upper Cretaceous volcanics of Maio, poles *A* and *B*, in comparison with the relevant polar curve for Africa and data from the Canary Islands. *B** is the *B*-pole after structural correction. Poles *C* and *D* represent the early subaerial volcanism (Series I) of Fuerteventura (Storetvedt *et al.* 1979), pole *E* is from lava Series I of North Lanzarote (Johansen 1976) and poles *F* and *G* are based on Upper Tertiary/Quaternary results from Fuerteventura and Gran Canaria/Tenerife (Storetvedt *et al.* 1978, 1979). In the inserted diagram crosses are African Tertiary poles and the numbers refer to the following data: (1) Mlanje (Briden 1967), (2) mean Mesozoic SE Africa (Hailwood & Mitchell 1971); (3) Lupata, 106 Myr (Gough & Opdyke 1963; Gough *et al.* 1964); (4) Shawa ijolite (Gough & Brock 1964); (5) mean Mesozoic NW Africa (Hailwood & Mitchell 1971); (6) Hoachanas (Gidskehaug, Creer & Mitchell 1975); (7) Kimberlite pipes, 83–89 Myr (McFadden & Jones 1977); (8) Wadi Natash volcanics, 81–90 Myr (El Shazly & Krs 1973); (9 and 10) L–(U) Tertiary volcanics Egypt (Grouda Hussain, Schult & Soffel 1979). Square symbols denote mean poles for: 1–6; 7 and 8; 9 and 10; and the crosses respectively (*cf.* inserted diagram).

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well into the Upper Cretaceous branch of the African polar wander path, agreeing particularly with data in the 90–70 Myr age range. Upon tilt correction the corresponding pole, B^* in Fig. 7, moves away from the inferred late Mesozoic polar track for Africa. Since the *B*-component is characterized by fairly shallow inclinations and the strike directions of the strata are generally close to the declination of this magnetization the structural correction has in general only a minor effect on the remanence dispersion. Only in locations where the strike of the strata varies from the general NNW direction do the corrected data differ greatly from the uncorrected ones. The overall effect of the tilt correction is that the precision parameter κ decreases from 24 to 13.5 and that α_{95} increases from 5.9° to 8.0°. Statistical comparison of the precisions (Watson 1956) also shows that they indeed are different at the required significance level ($p = 0.05$). This dispersion increase along with the somewhat discordant B^*-pole position suggests that the *B*-magnetization was acquired after the doming of the Central Igneous Complex, or possibly at a late stage of this process, in Upper Cretaceous time. The two-polarity build-up of this early magnetization suggests that the original cooling remanence (TRM) was replaced or strongly modified by late deuteric thermochemical processes that lasted at least for a sufficient length of time to record polarity inversion(s). Though the bulk of the investigated rocks come from a tectonically disturbed area the *B*-directions are very well grouped and there is apparently no systematic palaeomagnetic difference between geological units. This give support to the geological evidence that the compressive tectonism in the study area pre-dates the doming (Stillman *et al.* 1982) which in turn must be older than the major phase of sill intrusion.

As demonstrated in Fig. 7 the *A*-pole shows close correspondence with palaeopoles from the late Tertiary–Quaternary volcanic sequences of the Canary Islands (Storetvedt *et al.* 1978, 1979) and with data of similar ages from continental Africa. This agreement is further supported by the excellent match of the *A*-magnetization with the minor data base of Miocene/Pliocene lavas from Maio and Santiago (*cf.* Fig. 5c). The coexistence of the *A*- and *B*-magnetizations in most sites suggests, therefore, that the original late Cretaceous magnetization (the *B*-component) has become strongly overprinted by remanence acquired during the second major phase of Maio magmatism, i.e. in late Miocene–Pliocene time. Comparisons of the relative importance of the *A*- and *B*-magnetizations show that the *B*-component is more frequently extracted at higher levels in the sedimentary pile (the stratal repetition is ignored here since the *B*-magnetization post-dates this deformation phase) while the *A*-magnetization predominates at lower 'stratigraphic' levels. The apparent rapid increase of the *A*-component with depth indicates that there may exist a major late Tertiary intrusive body at a fairly shallow depth beneath the Jurassic sediments of East Maio but at present there is no further information available to support this suggestion.

Discussion and conclusions

The ample evidence for partial remagnetization appears to be closely connected with the substantial K/Ar age resetting of the Mesozoic volcanics of South and East Maio. The Batalha Formation, consisting of pillow lavas of Middle–Upper Jurassic age (*ca.* 150 Myr), show ages varying from 18 to 64 Myr while the inferred Upper Cretaceous volcanics, for example our site 70 which is regarded as a lava flow of Albian/Senomanian age, only gives ages around 10 Myr (Mitchell, private communication). It has also been previously suggested (Bernard-Griffiths *et al.* 1975) that a major Ar-degassing must have affected these rocks.

Other radiometric data that may be of interest for the age consideration on the island of Maio comes from the nearby DSDP sites 367 and 368 (Duncan & Jackson 1977). Site 368 is located on the Cape Verde Rise, north-east of the Cape Verde Islands. The drill bottomed in diabase intrusions interstratified with Upper Cretaceous black shales. Whole rock K/Ar

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ages of the basalt give ages slightly less than 20 Myr and so do Ar^{40}/Ar^{39} total fusion data. Step heating Ar^{40}/Ar^{39} results are more scattered and difficult to interpret but the majority of these dates also give similar ages. Thus there are reasons for believing that the site 368 intrusion corresponds to the early Neogene volcanism on Maio, correlating most likely with the Casas Velhas Formation that pre-dates the mid-Miocene Pedro Vaz sequence.

Hole 367 is located in the Cape Verde Basin, south-east of the Archipelago. The drilling ended in basalts overlying a slightly recrystallized Upper Jurassic limestone. Again, the basalt is most likely an intrusive rock though the thermal alteration of the adjacent sediment is much less pronounced than for site 368. Two whole-rock K/Ar dates give ages of 88 and 92 Myr. A total fusion Ar^{40}/Ar^{39} age is about 102 Myr but the corresponding incremental heating data give determined ages between 124 and 95 Myr, the ages decreasing fairly steadily from the low temperature fractions to complete fusion. On the whole an age of 90–100 Myr appears to be a reasonable age figure for the basalt of site 367, which is then most likely to be correlated with the Albian/Senomanian volcanic phase in Maio. There is no evidence of more than one igneous event at each of the two DSDP sites, referred to here, and in both cases the basalts are likely to represent intra-sedimentary intrusions. Consequently, there is no reason to suggest reheating of these basalts, a situation quite unlike the one encountered in the island of Maio.

We conclude that the major phase of sheet intrusions in the Albian–? Senomanian sedi ments of East Maio acquired their 'primary' magnetization in Upper Cretaceous time, probably in the 90–70 Myr age range. This intrusive 'event', which at least affected a sedi mentary pile of the order of a few hundred metres in thickness (we do not know to what extent Upper Cretaceous sediments have been stripped off by erosion), led to important secondary changes of the original (cooling) remanence. From the available mineralogical evidence and the two-polarity build-up of the Cretaceous magnetization there are good reasons for believing that this remanent magnetism is of low temperature origin, being consequently somewhat delayed compared to the actual time of intrusion. The results suggest that the main intrusive phase came after the domal uplift of the Central Igneous Complex which formed the stratal dips (as recorded in the surrounding sediments) away from this magmatic centre. In subsequent times the region of Maio was apparently dominated by erosion for perhaps a period of 70 Myr, until the Neogene volcanism began, probably in the early Miocene. This latter magmatism which led to rise in temperature and increase in chemical activity in the pre-existing rocks introduced a strong secondary magnetization over print, the importance of which increases with depth. Thus, in the study area there are reasons for assuming that a major late Neogene intrusive body exists at a fairly shallow depth. In East Maio the Miocene–Pliocene magnetization overprint seems to be associated with a practically complete radiometric age resetting. The available K/Ar dates from the Batalha Formation suggest that also the southern flank of the Central Igneous Complex has suffered a substantial loss of argon.

Important details of the tectonomagmatic evolution of Maio, such as: (1) major crustal uplift, local tectonism and volcanism at around Albian, (2) the following long time span of volcanic quiescence and erosion and (3) the final outbursts of volcanism at around Miocene, bear a close resemblance to the geological evolution of the Island of Fuerteventura in the Canaries (Storetvedt 1980).

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